Neehar Peri

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Education

Ph.D in Robotics, Carnegie Mellon University May 2026
Towards Scalable Open-World 3D Perception

M.S in Robotics, Carnegie Mellon University May 2023
Long-Tailed 3D Detection via Multi-Modal Fusion

B.S. in Computer Engineering, University of Maryland - College Park May 2021
QUEST Honors Program

Work Experience

California Institute of Technology, Pasadena, CA, *Postdoctoral Research Fellow* *Jun 2026 – Present*

- Leading research on vision-language action models and interactive simulators with video diffusion models for contact-rich manipulation
- Advisor: Georgia Gkioxari

RAI Institute, Cambridge, MA, *Research Scientist Intern* *Jun 2025 – May 2026*

- Benchmarked SLAM and SfM methods for UMI-style data collection with motion capture data
- Led research on policy learning with multi-modal representations (e.g. depth, force, tactile) for manipulation

Carnegie Mellon University, Pittsburgh, PA, *Graduate Research Assistant* *Apr 2020 – Apr 2026*

- Led research on 3D object detection, multi-object tracking, motion forecasting, and multi-agent planning for embodied perception
- Advisor: Deva Ramanan

NVIDIA, Pittsburgh, PA, *Research Scientist Intern* *Jan 2024 – Dec 2024*

- Led research on 3D object detection in-the-wild
- Built GNN-based tracker that outperforms production system by 5% HOTA and achieves a 10x speedup

MUKH Technologies, College Park, MD, *Research Scientist Intern* *Aug 2020 – May 2023*

- Led research on improving thermal-to-visible face synthesis for zero-shot identification
- Built robust face verification pipelines for multi-spectral data streams

Argo AI, Pittsburgh, PA, *Research Scientist Intern* *May 2021 – Oct 2022*

- Developed end-to-end 3D object detection and forecasting pipeline from LiDAR point clouds
- Implemented novel metrics that jointly evaluate detection and forecasting accuracy

University of Maryland, College Park, MD, *Undergraduate Research Assistant* *May 2018 – May 2021*

- Conducted research in unsupervised traffic anomaly detection and discriminative representation learning for vehicle re-id
- Led research in defending against clean-label adversarial poisoning attacks
- Established novel method for encoding human preferences in revenue maximizing auction design
- Advisors: Rama Chellappa & John P. Dickerson

Awards

Name	Institution	Distinction	Year
CVPR Doctoral Consortium	CMU	International	2026
NSF Graduate Research Fellowship	CMU	National	2023
Maryland Undergraduate Researcher of the Year	UMD	University	2021
Sujan Guha Memorial Best Senior Thesis Award	UMD	Department	2021
CRA Outstanding Undergraduate Researcher (Honorable Mention)	UMD	National	2021
Yurie & Jeong H. Kim Scholarship	UMD	Department	20{18,19,20}

Conference Publications

- RefAV: Towards Planning-Centric Scenario Mining — CVPR 2026††
  *C Davidson, D Ramanan, **N Peri***
- RF-DETR: Neural Architecture Search for Real-Time Detection Transformers — ICLR 2026
  *I Robinson, P Robicheaux, M Popov, D Ramanan, **N Peri***
- Roboflow100-VL: A Multi-Domain Object Detection Benchmark for Vision-Language Models — NeurIPS 2025
  P Robicheaux, M Popov*, A Madan, I Robinson, J Nelson, D Ramanan, **N Peri***
- MonoFusion: Sparse-View 4D Reconstruction via Monocular Fusion — ICCV 2025
  *Z Wang, J Tan, T Khurana⋆, **N Peri**⋆, D Ramanan*
- Towards Learning to Complete Anything in LiDAR — ICML 2025
  *A Tacmaz, C Saltori, **N Peri**, T Meinhardt, RD Lutio, L Leal-Taixe, A Osep*
- Planning with Adaptive World Models for Autonomous Driving — ICRA 2025
  *AB Vasudevan, **N Peri**, J Schneider, D Ramanan*
- Neural Eulerian Scene Flow Fields — ICLR 2025
  *K Vedder, **N Peri**, I Khatri, S Li, E Eaton, M Kocamaz, Y Wang, Z Yu, D Ramanan, J Pehserl*
- Revisiting Few-Shot Object Detection with Vision-Language Models — NeurIPS 2024
  A Madan, **N Peri***, S Kong, D Ramanan*
- Shelf-Supervised Cross-Modal Pre-Training for 3D Object Detection — CoRL 2024
  M Khurana, **N Peri***, J Hays, D Ramanan*
- I Can't Believe It's Not Scene Flow! — ECCV 2024
  I Khatri, K Vedder*, **N Peri**, D Ramanan, J Hays*
- Better Call SAL: Towards Learning to Segment Anything in LiDAR — ECCV 2024
  A Osep, T Meinhardt*, F Ferroni, **N Peri**, D Ramanan, L Leal-Taixe*
- ZeroFlow: Scalable Scene Flow via Distillation — ICLR 2024
  *K Vedder, **N Peri**, N Chodosh, I Khatri, E Eaton, D Jayaraman, Y Liu, D Ramanan, J Hays*
- Towards Long-Tailed 3D Detection — CoRL 2022
  ***N Peri**, A Dave, D Ramanan⋆, S Kong⋆*
- A Brief Survey of Person Recognition at a Distance — ASILOMAR 2022
  C Nalty, **N Peri***, J Gleason*, CD Castillo, S Hu, T Bourlai, R Chellappa*
- Forecasting from LiDAR via Future Object Detection — CVPR 2022
  ***N Peri**, J Luieten, M Li, A Osep, L Leal-Taixe, D Ramanan*
- Assessment of a Novel Virtual Environment for Examining Human Cognitive-Motor Performance during Execution of Action Sequences — HCII 2022
  AA Shaver, **N Peri***, R Mezebish, G Matthew, A Berson, C Gaskins, GP Davis, GE Katz, I Samuel, JA Reggia, J Purtilo, RJ Gentili*
- A Synthesis-Based Approach for Thermal-to-Visible Face Verification — FG 2021
  ***N Peri**, J Gleason, CD Castillo, T Bourlai, VM Patel, R Chellappa*
- PreferenceNet: Encoding Human Preferences in Auction Design with Deep Learning — NeurIPS 2021
  N Peri, MJ Curry*, S Dooley, JP Dickerson*
- The Devil is in the Details: Self-Supervised Attention for Vehicle Re-ID — ECCV 2020
  P Khorramshahi, **N Peri***, JC Chen, R Chellappa*
- A Dual Path Model with Adaptive Attention for Vehicle Re-ID — ICCV 2019††
  *P Khorramshahi, A Kumar, **N Peri**, SS Rambhatla, JC Chen, R Chellappa*

Journal Publications

- Accelerating Image Recognition Using High Performance Computing — ITEA 2023
  *J Adams, JM Barton, R Chellappa, J Gabberty, J Gleason, S Hu, J Johnson, F Moor-Clingenpeel, B Oshiro, **N Peri**, D Richie, V To*
- Data and Algorithms for End-to-End Thermal Spectrum Face Verification — TBIOM 2023
  *T Bourlai, J Rose, S Mokalla, A Zabin, L Hornak, CB Nalty, **N Peri**, J Gleason, CD Castillo, VM Patel, R Chellappa*

Workshop Publications

- QuickDraw: Fast Visualization, Analysis and Active Learning for Medical Image Segmentation — HCII 2025
  D Syomichev, P Gopinath*, GL Wei, E Chang, I Gordon, A Seifu, R Pemmaraju⋆, **N Peri**⋆, J Purtilo⋆*
- Semi-Supervised Federated Multi-Organ Segmentation with Partial Labels — AAPM 2024††
  R Pemmaraju, **N Peri****
- An Empirical Analysis of Range for 3D Object Detection — ICCV 2023††
  ***N Peri**, M Li, B Wilson, YX Wang, J Hays, D Ramanan*
- ReBound: An Open-Source 3D Bounding Box Annotation Tool for Active Learning — CHI 2023†
  W Chen, A Edgley*, R Hota*, J Liu*, E Schwartz*, A Yizar*, **N Peri**⋆, J Purtilo⋆*
- Deep k-NN Defense Against Clean-label Data Poisoning Attacks — ECCV 2020†
  N Peri*, *N Gupta*, WR Huang*, L Fowl, C Zhu, S Feizi, T Goldstein, JP Dickerson*
- Towards Real-Time Systems for Vehicle Re-ID, Multi-Camera Tracking, and Anomaly Detection — CVPR 2020†
  N Peri*, *P Khorramshahi*, SS Rambhatla*, V Shenoy, S Rawat, JC Chen, R Chellappa*
- Attention Driven Vehicle Re-ID and Unsupervised Anomaly Detection for Traffic Understanding — CVPR 2019†
  *P Khorramshahi, **N Peri**, A Kumar, A Shah, R Chellappa*

Preprints

- Bridging the Embodiment Gap: Jointly Training on UMI and Teleoperated Data Improves Cross-Robot Policy Transfer — Under Review
  V Surendran, **N Peri****, *D Watkins*
- MimicAgent: Learning Quadruped Skills via Text-to-Trajectory Generation — Under Review
  NP Parameswaran, LK Nayak*, **N Peri**, D Ramanan*
- DetPO: In-Context Learning with Multi-Modal LLMs for Few-Shot Object Detection — Under Review
  GR Gare, **N Peri****, *M Popov, SP Jain, J Galeotti, D Ramanan*
- UniFlow: Zero-Shot LiDAR Scene Flow for Autonomous Driving — Under Review
  *S Li, Q Zhang, I Khatri, K Vedder, E Eaton, D Ramanan, **N Peri***
- A Reality Check for Open-Vocabulary Monocular 3D Object Detection — Under Review
  *M Gladkova, I Khatri, D Ramanan, D Cremers, **N Peri***
- Long-Tailed 3D Detection via Multi-Modal Late Fusion — Under Review
  Y Ma, **N Peri****, *A Dave, W Hua, D Ramanan, S Kong*

*Equal Contribution
⋆Equal Supervision
†Selected for Spotlight Presentation
††Selected for Oral Presentation

Invited Talks

- Foundational Few-Shot Object Detection Challenge — Jun 2026
  Invited Talk: CVPR 2026, Workshop on Open World Vision
- Argoverse 2 Scenario Mining Challenge — Jun 2025
  Invited Talk: CVPR 2025, Workshop on Autonomous Driving
- Foundational Few-Shot Object Detection Challenge — Jun 2025
  Invited Talk: CVPR 2025, Workshop on Visual Perception via Learning in an Open World
- 3D Object Detection for Autonomous Vehicles — Apr 2025
  Guest Lecture: 16-825, Learning for 3D Vision
- Towards Foundation Models for 3D Perception — Mar 2025
  Invited Talk: Carnegie Mellon University (FLAME Seminar & NeuroAI Seminar)
- Image Processing from a Frequency Perspective — Feb 2025
  Guest Lecture: 16-720, Computer Vision
- Long-Tailed 3D Detection via 2D Late Fusion — Oct 2024
  Invited Talk: ECCV 2024, Workshop on Vision-Centric Autonomous Driving
- Shelf-Supervised Cross-Modal Pre-Training for 3D Object Detection — Oct 2024
  Invited Talk: ECCV 2024, Autonomous Vehicles meet Multimodal Foundation Models Workshop
- Argoverse 2 End-to-End Forecasting Challenge — Jun 2024
  Invited Talk: CVPR 2024, Workshop on Autonomous Driving
- Foundational Few-Shot Object Detection Challenge — Jun 2024
  Invited Talk: CVPR 2024, Workshop on Visual Perception via Learning in an Open World

- 3D Object Detection for Autonomous Vehicles — Apr 2024
  Guest Lecture: 16-720, Computer Vision
- Better Call SAL: Towards Learning to Segment Anything in LiDAR — Apr 2024
  Invited Talk: Stack AV
- 3D Object Detection for Autonomous Vehicles — Apr 2024
  Guest Lecture: 16-825, Learning for 3D Vision
- Long-Tailed 3D Object Detection via Multi-Modal Fusion — Jan 2024
  Invited Talk: Carnegie Mellon University (R-PAD Lab)
- An Empirical Analysis of Range for 3D Object Detection — Oct 2023
  Invited Talk: ICCV 2023, Robustness and Reliability of Autonomous Vehicles in the Open-World
- Argoverse 2 End-to-End Forecasting Challenge — Jun 2023
  Invited Talk: CVPR 2023, Workshop on Autonomous Driving
- 3D Object Detection for Autonomous Vehicles — Mar 2023
  Guest Lecture: 16-825, Learning for 3D Vision
- Image Processing and Convolutions — Sep 2022
  Guest Lecture: 16-720, Computer Vision
- How do Autonomous Vehicles See the World? — Aug 2022
  Invited Talk: Carnegie Mellon University (RoboLaunch)
- Transformers for Vision — Apr 2022
  Guest Lecture: 16-720, Computer Vision
- Training Convolutional Neural Networks — Apr 2022
  Guest Lecture: 16-720, Computer Vision
- Metrics and Methods for Detection and Forecasting in Autonomous Vehicles — Apr 2022
  Invited Talk: National Autonomous Vehicle Conference

Mentorship

Name	Institution	Year(s)	Project
Narayanan Parameswaran	CMU	2025 –	Text-to-policy for quadrupedal locomotion
Lucky Nayak	CMU	2025 –	Text-to-policy for quadrupedal locomotion
Shruti Jain	CMU	2025	Benchmarking spatial understanding in VLMs
Siyi Li	UPenn	2025 –	Zero-shot scene flow estimation via cross-domain generalization, VLM post-training for spatial understanding
Cainan Davidson	CMU	2024 –	Benchmarking scenario mining for AVs, VLAs for offroad driving
Guang-Lin Wei, Eric Chang, Padmini Gopinath, Ian Gordon, Amanuel Seifu, Daniel Syomichev	UMD	2024	CMSC435 software engineering capstone to build an active-learning framework for medical image analysis
Zihan Wang	CMU → Berkeley PhD	2024 – 2025	Sparse-view dynamic reconstruction in-the-wild
Nina Johe, Aryan Kakadia, Muzzamil Khan, Morgan Ko, Josh Leeman, Max Son, Sashwat Venkatesh	UMD	2024	CMSC435 software engineering capstone to build an end-to-end platform for medical image analysis
Mehar Khurana	IIITD → CMU MSR	2023 – 2024	Shelf-supervised 3D object detection with VLMs
Anish Madan	CMU → SkildAI	2022 – 2025	Few-shot multi-modal 2D detection with VLMs
Andrew Shen	CMU → OpenAI	2022 – 2023	Benchmarking modular 3D perception stack for AVs
Aminah Yizar, Andrew Edgley, Ezra Schwartz, Joshua Liu, Raunak Hota, Royce He, Wesley Chen	UMD	2022	CMSC435 software engineering capstone to build an active learning framework to allow human-in-the-loop 3D object annotation

Christopher Nalty	MUKH → Oregon State PhD	2021 – 2022	Synthetic data augmentation for thermal-to-visible face verification
Aastha Senjalia, Andrew Vetter, Benjamin Namovicz, Cheyenne Montgomery, Ferzam Mohammad, Matthew Weinberg, Nicholas Revill	UMD	2021	CMSC435 software engineering capstone to build a visualization platform for autonomous vehicle data. Project won People's Choice Award.

Professional Service

Conference Reviewer: NeurIPS, CVPR, AAAI, ICCV, ICLR, ECCV, ICRA
Journal Reviewer: IJCV, TPAMI
Mentorship: CMU AI Mentoring Program, QUEST Mentoring Program, CMU AI for Social Good Summit
Masters Thesis Committee Member: Bharath Raj, Anish Madan
Other: AUTOBOT & TRINITY Cluster Management, RISS Admissions Committee, CMU Computer Vision Reading Group

Organized Workshops & Challenges

Event	Venue	Year
6th Workshop on Open-World Vision (OWV)	CVPR	2026
RF20-VL Few-Shot Object Detection Challenge @ OWV Workshop	CVPR	2026
Scenario Mining Challenge @ Workshop on Autonomous Driving	CVPR	2026
Scene Flow Challenge @ Workshop on Autonomous Driving	CVPR	2026
5th Workshop on Visual Perception and Learning in an Open World (VPLOW)	CVPR	2025
RF20-VL Few-Shot Object Detection Challenge @ VPLOW Workshop	CVPR	2025
Scenario Mining Challenge @ Workshop on Autonomous Driving	CVPR	2025
4th Workshop on Visual Perception and Learning in an Open World (VPLOW)	CVPR	2024
nuImages Few-Shot Object Detection Challenge @ VPLOW Workshop	CVPR	2024
End-to-End Forecasting Challenge @ Workshop on Autonomous Driving	CVPR	2024
3rd Workshop on Visual Perception and Learning in an Open World (VPLOW)	CVPR	2023
End-to-End Forecasting Challenge @ Workshop on Autonomous Driving	CVPR	2023
2nd Workshop on Visual Perception and Learning in an Open World (VPLOW)	CVPR	2022

Teaching Experience

16-720, Carnegie Mellon University, Robotics Institute, *Head Teaching Assistant* — *Spring 2022, Fall 2022*
- Managed team of teaching assistants to effectively coordinate course responsibilities
- Graded course projects and held office hours

ENEE 244, University of Maryland, ECE Department, *Undergraduate Teaching Fellow* — *Spring 2019*
- Led Introduction to Digital Logic recitation for a discussion section of 15 students